

Solutions to Problem Solving Class 1: Error Detection, Coding, and Modulation

Computer Communications – DE
Department of Communication Systems, LTH
ht 2006

1. There should be an even number of ones in the message. Therefore, the parity bits are as follows:
 - a) 0
 - b) 0
 - c) 1
 - d) 0
2. The system is using even parity, and there are five positions equal to 1. Thus the pattern is in error.
3. If the data rate is X bps then each bit lasts 1/X seconds.
 - a) 3 bits
 - b) 24 bits
 - c) 192 bits

4. $x^8 + x^3 + x + 1 \leftrightarrow 100001011$.

5. $100001110001 \leftrightarrow x^{11} + x^6 + x^5 + x^4 + 1$

6. Sequence after adding $n = 3$ extra zeros: 1010011110000. Divisor (of length $n + 1$): 1011. The polynomial division is

$$\frac{x^{12} + x^{10} + x^7 + x^6 + x^5 + x^4}{x^3 + x + 1} = x^9 + x^6 + x^2 + x + 1 + \frac{1}{x^3 + x + 1}.$$

The remainder polynomial is 1, so CRC=001.

7. Sequence after adding remainder: 10110011111. Divisor: 1001. The polynomial division is

$$\frac{x^{10} + x^8 + x^7 + x^4 + x^3 + x^2 + x + 1}{x^3 + 1} = x^7 + x^5 + x^2 + x + 1.$$

The remainder polynomial is 0, so there is no error.

8. $M=FF01D_{16}=11111111000000011101_2$. $P=33_{16}=00110011_2$. The polynomial division is

$$\begin{aligned} \frac{x^{19} + x^{18} + x^{17} + x^{16} + x^{15} + x^{14} + x^{13} + x^{12} + x^4 + x^3 + x^2 + 1}{x^5 + x^4 + x + 1} &= \\ &= x^{14} + x^{12} + \frac{x^4 + x^3 + x^2 + 1}{x^5 + x^4 + x + 1}. \end{aligned}$$

The remainder polynomial is $x^4 + x^3 + x^2 + 1 \neq 0$, so there is an error.

9. $M=3AB_{16}=001110101011_2$. $P=1B_{16}=00011011_2$. We have that $n+1=5$, so the data has to be extended by $n=4$ zeros. The polynomial division is

$$\frac{x^{13} + x^{12} + x^{11} + x^9 + x^7 + x^5 + x^4}{x^4 + x^3 + x + 1} = x^9 + x^7 + x^4 + x^3 + x^2 + x + \frac{x}{x^4 + x^3 + x + 1}.$$

The remainder polynomial is x , so CRC=0010.

10. a)

$$\frac{x^7 + x^6 + x^4 + x^2 + x + 1}{x^4 + x^3 + 1} = x^3 + 1 + \frac{x^2 + x}{x^4 + x^3 + 1}$$

remainder $\neq 0$, so no acceptance.

b)

$$\frac{x^{10} + x^8 + x^6 + x^5 + x^3 + x^2 + 1}{x^4 + x^3 + 1} = x^6 + x^5 + \frac{x^3 + x^2 + 1}{x^4 + x^3 + 1}$$

remainder $\neq 0$, so no acceptance.

c)

$$\frac{x^{10} + x^6 + x^5 + x^4 + x^2 + x + 1}{x^4 + x^3 + 1} = x^6 + x^5 + x^4 + x^3 + x^2 + x + 1$$

remainder = 0, so acceptance.

11. The sum is 11111111.

12. The checksum is 1101010000011110.

13. The message is correct if the complement of the sum is 0.

a) No

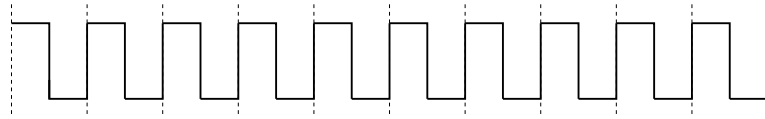
b) Yes

c) Yes

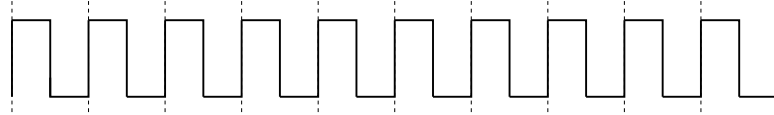
14. a)

[illegible]

b)



c)



15. The sequence is:

a) 0 1 0 0 1 1 0 0 0 1

b) 1 1 0 0 1 1 0 1 1 0

c) 1 0 0 0 0 1 1 1 0 0

16. The sequence is:

a) 1 1 1 0 1 0 1 0 0 1

b) 1 0 1 0 1 0 1 1 0 1

c) 1 1 0 0 0 1 0 0 1 0